Topic 2: Programming

Learning to program is a core component of a computer science course. Students should be competent at designing, reading, writing and debugging programs. They must be able to apply their skills to solve real problems and produce robust programs.

Students should be given repeated opportunities to develop and practise their programming skills. They will develop their knowledge and skills in at least one of the following programming languages:

1. Python
2. C#
3. Java.

	Students should:
2.1 Develop code	**2.1.1** Be able to write programs in a high-level programming language. **2.1.2** Understand the benefit of producing programs that are easy to read and be able to use techniques (comments, descriptive names (variables, constants, subprograms), indentation) to improve readability and to explain how the code works. **2.1.3** Be able to differentiate between types of error in programs (logic, syntax, runtime). **2.1.4** Be able to design and use test plans and test data (normal, boundary, erroneous). **2.1.5** Be able to interpret error messages and identify, locate and fix errors in a program. **2.1.6** Be able to determine what value a variable will hold at a given point in a program (trace table). **2.1.7** Be able to determine the strengths and weaknesses of a program and suggest improvements.